

Circuit Theory

Chapter 6

Capacitors and Inductors

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Capacitors and Inductors

Chapter 6

6.1 Capacitors

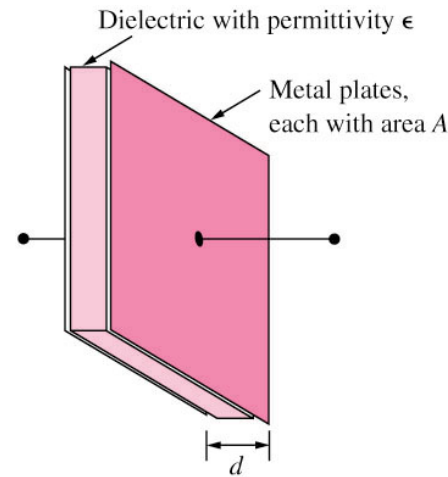
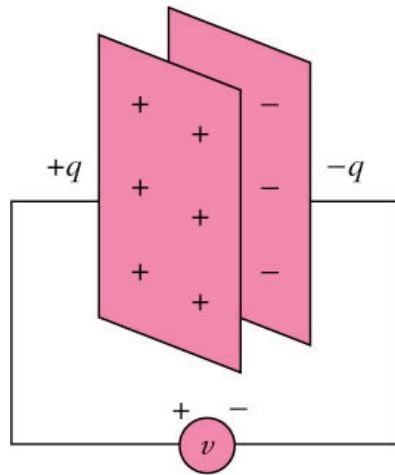
6.2 Series and Parallel Capacitors

6.3 Inductors

6.4 Series and Parallel Inductors

6.1 Capacitors (1)

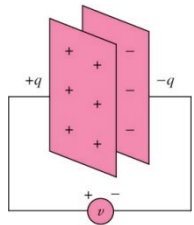
- A capacitor is a passive element designed to **store energy** in its **electric field**.



- A **capacitor** consists of two conducting plates separated by an insulator (or dielectric).

6.1 Capacitors (2)

- **Capacitance** C is the ratio of the charge q on one plate of a capacitor to the voltage difference v between the two plates, measured in farads (F).



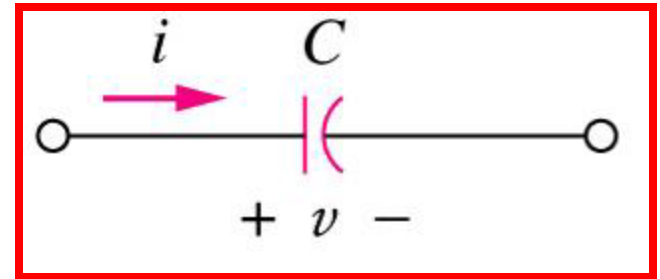
$$q = C v \quad \text{and}$$

$$C = \frac{\epsilon A}{d}$$

- Where ϵ is the permittivity of the dielectric material between the plates, A is the surface area of each plate, d is the distance between the plates.
- Unit: F, pF (10^{-12}), nF (10^{-9}), and μF (10^{-6})

6.1 Capacitors (3)

- If i is flowing into the +ve terminal of C
 - Charging $\Rightarrow i$ is +ve
 - Discharging $\Rightarrow i$ is -ve



- The current-voltage relationship of capacitor according to above convention is

$$i = C \frac{d v}{d t}$$

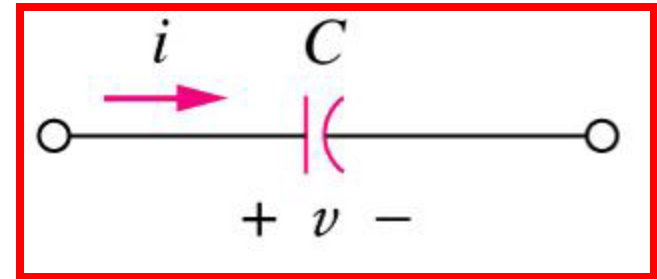
and

$$v = \frac{1}{C} \int_{t_0}^t i d t + v(t_0)$$

6.1 Capacitors (4)

- The energy, **w**, stored in the capacitor is

$$w = \frac{1}{2} C v^2$$



- A capacitor is
 - an **open circuit** to dc ($dv/dt = 0$).
 - its voltage **cannot change abruptly**.

6.1 Capacitors (5)

Example 1

The current through a 100- μ F capacitor is

$$i(t) = 50 \sin(120 \pi t) \text{ mA.}$$

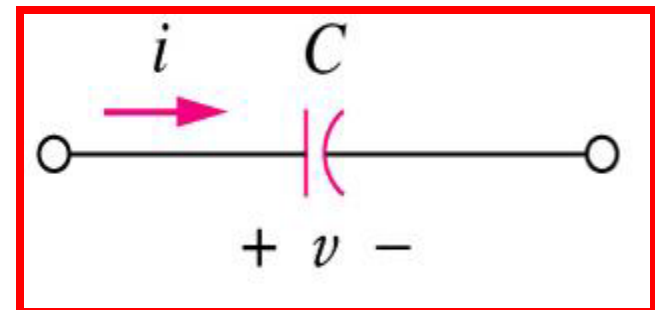
Calculate the voltage across it at $t = 1 \text{ ms}$ and $t = 5 \text{ ms}$.

Take $v(0) = 0$.

Answer:

$$v(1\text{ms}) = 93.14\text{mV}$$

$$v(5\text{ms}) = 1.7361\text{V}$$

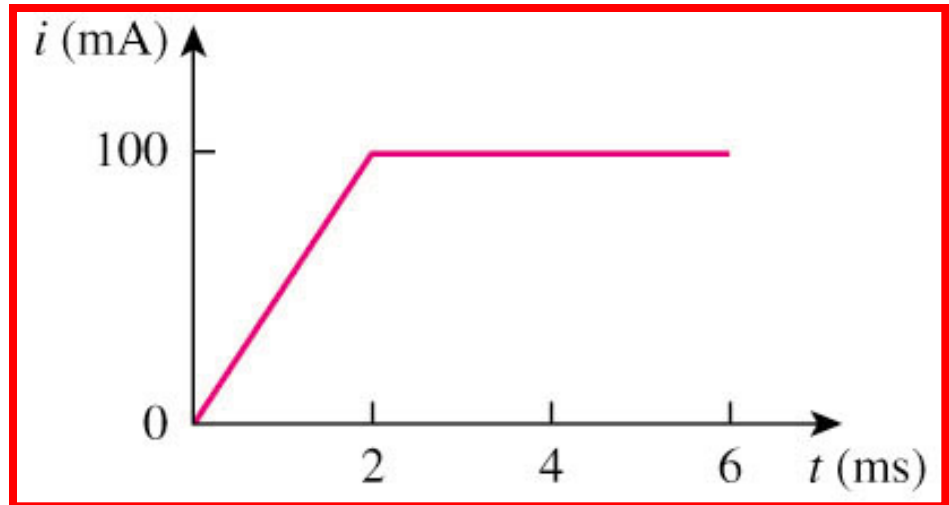


6.1 Capacitors (6)

Example 2

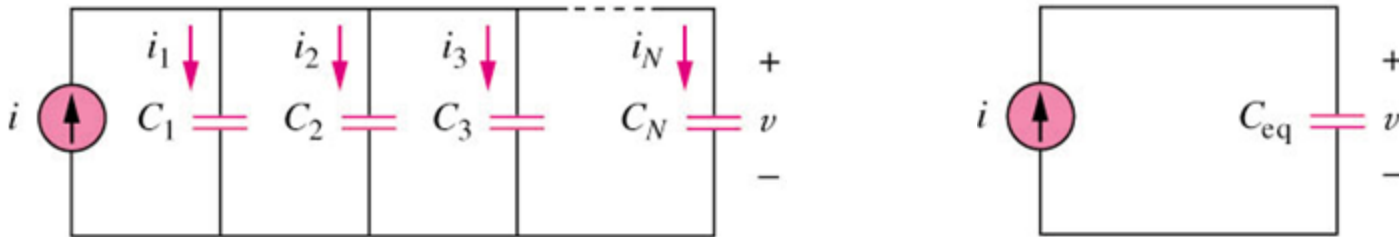
An initially uncharged 1-mF capacitor has the current shown below across it.

Calculate the voltage across it at $t = 2$ ms and $t = 5$ ms.



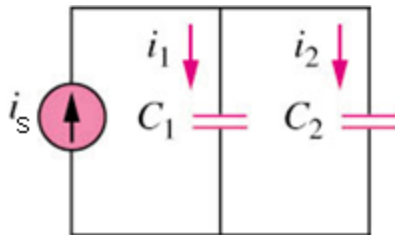
6.2 Series and Parallel Capacitors (1)

- The equivalent capacitance of N **parallel-connected** capacitors is the sum of the individual capacitances.



$$C_{eq} = C_1 + C_2 + \dots + C_N$$

- Current Divider Rule

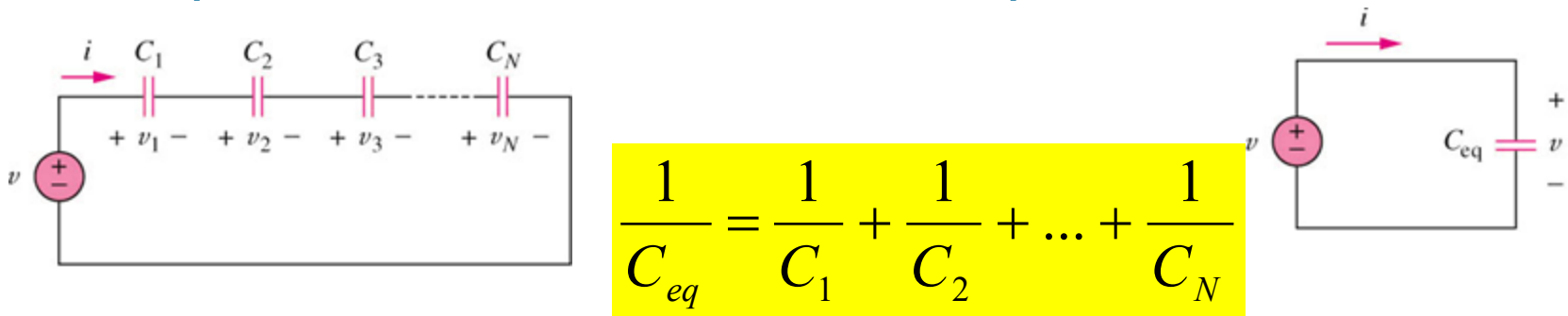


$$i_1 = \frac{C_1}{C_1 + C_2} i_s$$

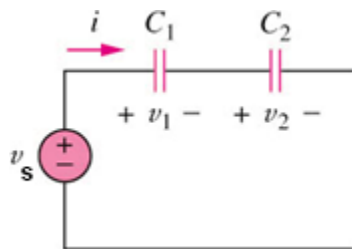
$$i_2 = \frac{C_2}{C_1 + C_2} i_s$$

6.2 Series and Parallel Capacitors (2)

- The equivalent capacitance of N **series-connected** capacitors is the reciprocal of the sum of the reciprocals of the individual capacitances.



- Voltage Divider Rule



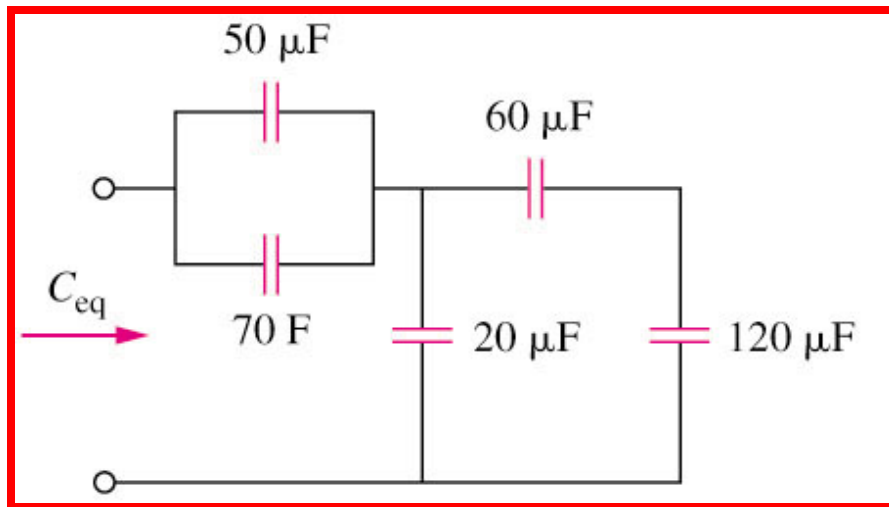
$$v_1 = \frac{C_2}{C_1 + C_2} v_s$$

$$v_2 = \frac{C_1}{C_1 + C_2} v_s$$

6.2 Series and Parallel Capacitors (3)

Example 3

Find the equivalent capacitance seen at the terminals of the circuit in the circuit shown below:



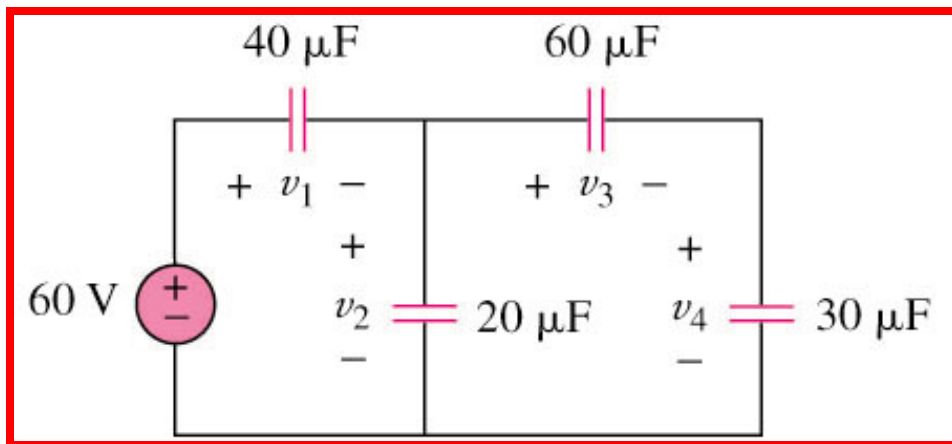
Answer:

$$C_{eq} = \underline{40\ \mu\text{F}}$$

6.2 Series and Parallel Capacitors (4)

Example 4

Find the voltage across each of the capacitors in the circuit shown below:



Answer:

$$v_1 = 30\text{V}$$

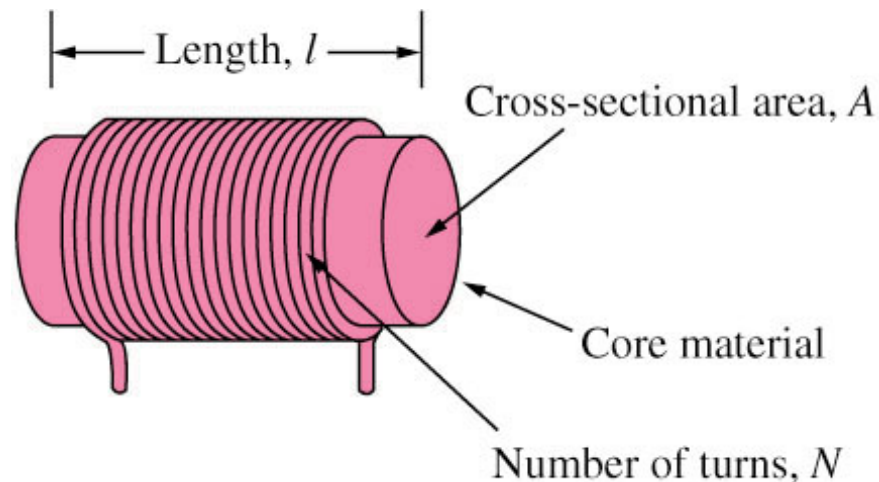
$$v_2 = 30\text{V}$$

$$v_3 = 10\text{V}$$

$$v_4 = 20\text{V}$$

6.3 Inductors (1)

- An inductor is a passive element designed to store energy in its magnetic field.



- An inductor consists of a coil of conducting wire.

6.3 Inductors (2)

- Inductance is the property whereby an inductor exhibits opposition to the change of current flowing through it, measured in henrys (H).

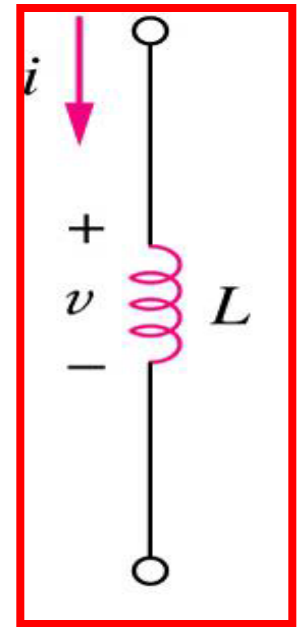
$$v = L \frac{d i}{d t} \quad \text{and} \quad L = \frac{N^2 \mu A}{l}$$

- μ is the inductor core permeability
- The unit of inductors is Henry (H), mH (10^{-3}) and μH (10^{-6}).

6.3 Inductors (3)

- The current-voltage relationship of an inductor:

$$i = \frac{1}{L} \int_{t_0}^t v(t) dt + i(t_0)$$



- The energy stored by an inductor:

$$w = \frac{1}{2} L i^2$$

- An inductor acts like a short circuit to dc ($di/dt = 0$) and its current cannot change abruptly.

6.3 Inductors (4)

Example 5

The terminal voltage of a 2-H inductor is

$$v = 10(1-t) \text{ V}$$

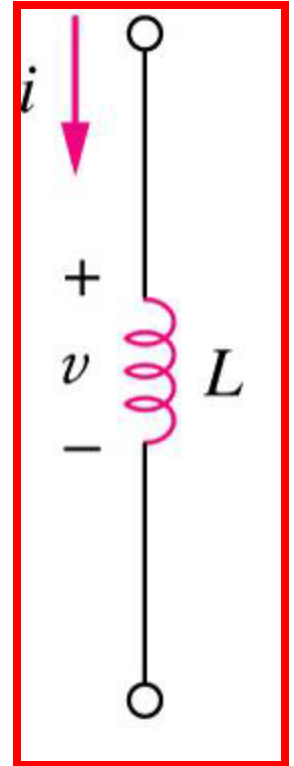
Find the current flowing through it at $t = 4 \text{ s}$ and the energy stored in it within $0 < t < 4 \text{ s}$.

Assume $i(0) = 2 \text{ A}$.

Answer:

$$i(4\text{s}) = -18\text{A}$$

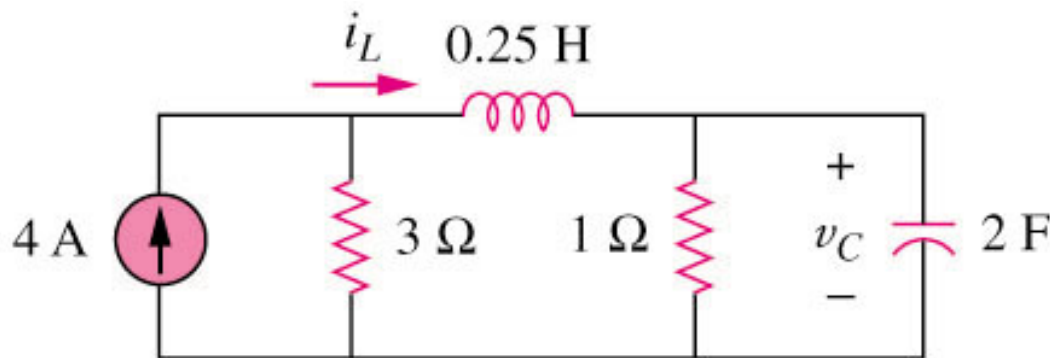
$$w(4\text{s}) = 320\text{J}$$



6.3 Inductors (5)

Example 6

Determine v_C , i_L , and the energy stored in the capacitor and inductor in the circuit shown below under dc conditions.



Answer:

$$i_L = 3A$$

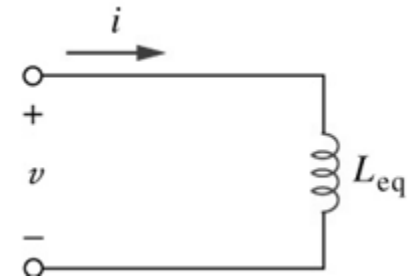
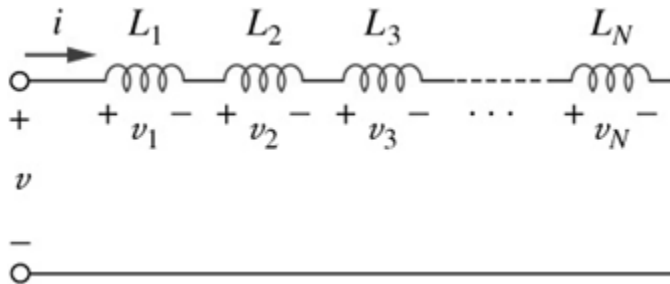
$$v_C = 3V$$

$$w_L = 1.125J$$

$$w_C = 9J$$

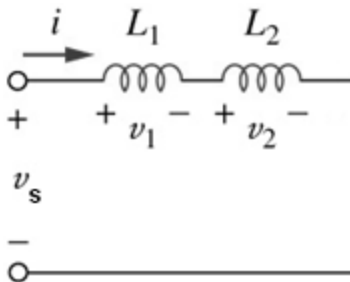
6.4 Series and Parallel Inductors (1)

- The equivalent inductance of **series-connected** inductors is the sum of the individual inductances.



$$L_{eq} = L_1 + L_2 + \dots + L_N$$

- Voltage Divider Rule

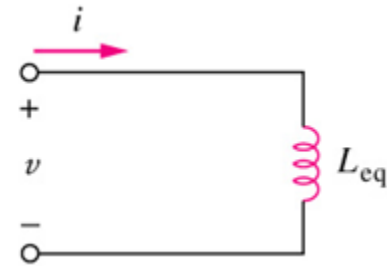
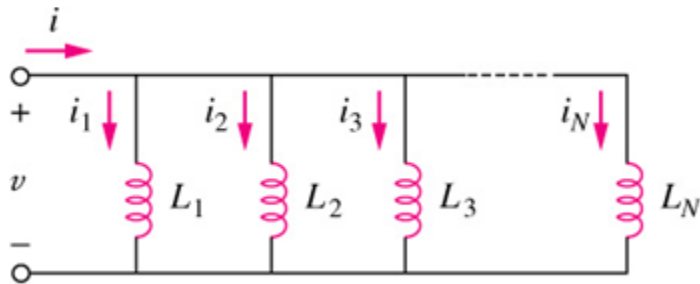


$$v_1 = \frac{L_1}{L_1 + L_2} v_s$$

$$v_2 = \frac{L_2}{L_1 + L_2} v_s$$

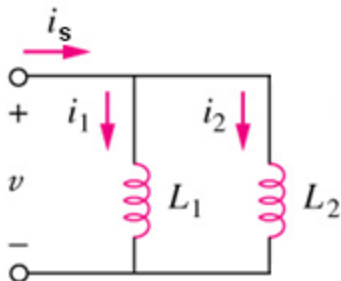
6.4 Series and Parallel Inductors (2)

- The equivalent capacitance of **parallel** inductors is the reciprocal of the sum of the reciprocals of the individual inductances.



$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \dots + \frac{1}{L_N}$$

- Current Divider Rule



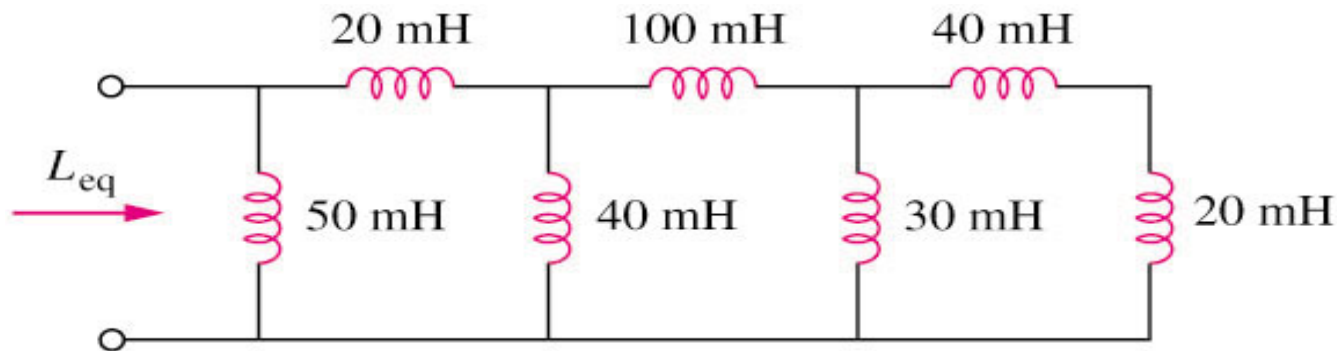
$$i_1 = \frac{L_2}{L_1 + L_2} i_s$$

$$i_2 = \frac{L_1}{L_1 + L_2} i_s$$

6.4 Series and Parallel Inductor (3)

Example 7

Calculate the equivalent inductance for the inductive ladder network in the circuit shown below:



Answer:

$$L_{eq} = \underline{25\text{mH}}$$

HW6 Ch6: 9, 13, 32, 61, 65, 73, 79

6.4 Current and voltage relationship

- Current and voltage relationship for R, L, C

Circuit element	Units	Voltage	Current	Power
 Resistance	ohms (Ω)	$v = Ri$ (Ohm's law)	$i = \frac{v}{R}$	$p = vi = i^2 R$
 Inductance	henries (H)	$v = L \frac{di}{dt}$	$i = \frac{1}{L} \int v dt + k_1$	$p = vi = Li \frac{di}{dt}$
 Capacitance	farads (F)	$v = \frac{1}{C} \int i dt + k_2$	$i = C \frac{dv}{dt}$	$p = vi = Cv \frac{dv}{dt}$